

Mario Andres Munoz Acosta

Curriculum Vitae

September 2026

📍 School of Computing and Information Systems
The University of Melbourne
🏠 <https://andremun.github.io/>
☎ +61 3 9035 9959
✉ munoz.m@unimelb.edu.au
🌐 andremun

Education and qualifications

2025 **Grad.Cert. in University Teaching** The University of Melbourne, Australia.
2009-2014 **Ph.D.** The University of Melbourne, Australia.
2012 **Grad.Cert. in Advanced Learning and Leadership** The University of Melbourne, Australia.
2006-2008 **M.Eng. (Electronics)** Universidad del Valle, Cali, Colombia.
1999-2005 **B.Eng. (Electronics)** Universidad del Valle, Cali, Colombia.

Employment history

2024- **Senior Research Fellow (Level C)** ARC Centre in Optimisation Technologies, Integrated Methodologies and Applications (OPTIMA), School of Computing and Information Systems, The University of Melbourne.
2022-2024 **Research Fellow (Level B)** ARC Centre in Optimisation Technologies, Integrated Methodologies and Applications (OPTIMA), School of Computing and Information Systems, The University of Melbourne.
2018-2019 **Lecturer** (0.2FTE) School of Computing and Information Systems, The University of Melbourne.
2017-2021 **Research Fellow (Level B)** School of Mathematics and Statistics, The University of Melbourne.
2014-2017 **Research Fellow (Level A)** School of Mathematical Sciences, Monash University.
2008-2009 **Auxiliary Professor** School of Electrical and Electronics Engineering, Universidad del Valle, Colombia.
2008 **Intern** Department of Electrical and Computer Engineering, University of Delaware, USA.

Research and Industry Engagement

- My research interests are randomised optimisation heuristics, automated optimisation and machine learning, operations research, experimental algorithmics, and modelling and simulation.
- I have authored 45 journal articles and 30 conference papers, which have gathered 2385 citations in Google Scholar (1397 in Scopus), resulting in an h-index of 25 (19), with 1651 (1027) citations over the last five years on an upward trajectory.
- I have worked with the following industry partners as a researcher or supervising student teams:
 - Future Fibre Technologies Ltd. (2016-2019): Development of methods for detecting intrusion events across various operating environments, while mitigating nuisances.
 - Powerline Bushfire Safety Program (2017): Algorithms to identify which plant species are likely to cause a fault when they come into contact with a 22 kV power line.
 - Boeing Australia (2021-2024): Methodologies that use multi-fidelity information to model an aircraft flight envelope.
 - IntelliSchool (2022): Development of models for identifying groups of students based on their academic development.
 - HatchQuarter (2022): Development of a model to predict demand for milk derivatives in global markets.
 - AGL Ltd. (2024): Development of an Anomaly Detection Model for Predictive Maintenance on Wind Turbines.

Service and other professional activities (Selected)

- I am member of the Health Analytics, Leadership, and Economics Research Hub, the Critical Minerals Consortium and the Australia-India Critical Minerals Research Hub.
- I am a member of the Education and Training committee at the ARC Centre in Optimisation Technologies, Integrated Methodologies and Applications (OPTIMA).
- I hold roles in conference organisation, including Hybridisation (Visualisation) co-chair (2024-2026) and the L4EC Track co-chair (2026,2027) for the flagship GECCO conferences, and Program committee for OPTIMA-CON (2024,2027), also serving as co-chair for the committee in 2024.
- I regularly serve on the program committee for flagship conferences, including AAAI, GECCO, and PPSN.
- I am a regular reviewer for, among others, the following Q1 Journals: (a) European Journal of Operational Research (b) Computers and Operations Research (c) Journal of Artificial Intelligence Research (d) IEEE Transactions on Cybernetics (e) IEEE Transactions on Evolutionary Computation (f) Evolutionary Computation (g) ACM Transactions on Evolutionary Learning and Optimisation (h) Swarm and Evolutionary Computation, and (i) Knowledge-Based Systems.

Teaching (Selected)

- COMP90083: *Computational Modelling and Simulation* (PG, Elective) School of Computing and Information Systems, The University of Melbourne.
 - **Sole Coordinator** [S2/2025] (about 50 students per session)
 - **Coordinator** and **Lecturer** of twelve sessions [S2/2023-2024] (about 40 students per session)
 - **Lecturer** of twelve sessions [S2/2022] (about 40 students per session)
- COMP90038: *Algorithms and Complexity* (PG, Compulsory) School of Computing and Information Systems, The University of Melbourne.
 - **Lecturer** of twelve sessions [S2/2018-2019] (about 350 students per session)
- ECE3093: *Optimisation, Estimation, and Numerical Methods* (UG, Compulsory), School of Mathematical Sciences, Monash University.
 - **Tutor** [S1/2017, S1/2016, S1/2015] (about 25 students per tutorial)

Honours and awards (Selected)

- 2019 *Finalist, Vienna Research Groups.* Vienna Science and Technology Fund, Austria.
- 2017 *Second place, Vegetation Detection Challenge.* Department of Environment, Land, Water and Planning, State of Victoria.

Research income (Selected)

- 2025 **M.A. Muñoz.** 2025DYA013 – *Leveraging Machine Learning and Ultra-Scale Computing for Solving Large Optimisation Problems.* Dyason Fellowships, The University of Melbourne (AU\$7,896)
- 2025 M. Czyz, A. Polyzos, K. Smith-Miles and **M.A. Muñoz.** IG240100160 – *A Scalable Machine Learning Platform for Pharmaceutical Manufacture.* Australia Economic Accelerator Ignite, Australia (AU\$428,296)
- 2024 C. Doerr, M.L. Santori, M. Gallagher, K. Smith-Miles and **M.A. Muñoz.** *Fitness landscapes of High-Dimensional Black-Box Optimisation in Structural Mechanics.* Franco-Australian Hubert Curien Program, France. (€4,000 – AU\$6,800)
- 2021 **M.A. Muñoz** and S. Kandanaarachchi. *Exploring strategies for constructing algorithm portfolios.* ARC Centre of Excellence for Mathematical and Statistical Frontiers Research Sprint Scheme, Australia. (AU\$20,000)
- 2018 M. Yellishetty, S. Northey, Y. Yuan, Z. Weng, S. Walsh and **M.A. Muñoz.** *Evaluating Critical Raw Material Supply for Australia's Defence Sector.* Defence Science Institute, Australia. (AU\$50,000)
- 2017 S. Kandanaarachchi, **M.A. Muñoz** and P.D. Talagala. *Preventing bushfires by early detection of branches on powerlines.* ARC Centre of Excellence for Mathematical and Statistical Frontiers Research Support Scheme, Australia. (AU\$13,468)
- 2017 K.A. Smith-Miles, R. Hyndman, **M.A. Muñoz** and J. Katsifolis. LP160101885 – *Intruder alert! Detecting and Classifying Events in Noisy Time Series.* Australian Research Council, Australia. (AU\$204,000)
- 2016 K.A. Smith-Miles, **M.A. Muñoz** and S. Kandanaarachchi. RC53128 – *Improved Intrusion Detection Analysis.* Department of Industry, Innovation and Science, Australia. (AU\$100,000)
- 2015 K.A. Smith-Miles and **M.A. Muñoz.** RC48547 – *New mathematical models for data handling.* Department of Industry, Innovation and Science, Australia. (AU\$100,000)

Invited talks

- 2024 *Tutorial – Instance Space Analysis for Rigorous and Insightful Algorithm Testing.* With K. Smith-Miles. PPSN'24.
- 2024 *Tutorial – Instance Space Analysis for Rigorous and Insightful Algorithm Testing.* With K. Smith-Miles. IJCNN'24.
- 2024 *Tutorial – Instance Space Analysis and Item Response Theory for Algorithm Testing.* With K. Smith-Miles and S. Kandanaarachchi. GECCO'24.
- 2023 *Tutorial – Instance Space Analysis for Rigorous and Insightful Algorithm Testing.* With K. Smith-Miles. PRICAL'23.
- 2023 *A few open questions related to automated algorithm selection.* Dagstuhl Seminar 23332 – Synergizing Theory and Practice of Automated Algorithm Design for Optimization.
- 2023 *Distribution Network Topology Identification – Generating multiple instances for algorithm benchmarking.* Workshop on Good Benchmarking Practices for Evolutionary Computation. GECCO'23.
- 2020 *Three questions on benchmarking, problem analysis and algorithm selection.* Lorentz Center Workshop – Benchmarked: Optimization meets Machine Learning.

- 2020 *Tutorial – Instance Space Analysis for Rigorous and Insightful Algorithm Testing*. With K. Smith-Miles. CEC'20.
- 2017 *Complexity and learning: How computers help us to gather insights from a messy world*. Geosciences Australia.
- 2017 *Workshop – Experiences publishing in Q1 Journals*. Universidad Militar Nueva Granada, Colombia.
- 2016 *Building and exploiting a non-parametric problem space*. Dagstuhl Seminar 16412 – Automated Algorithm Selection and Configuration.

Researcher supervision

Current

Doctoral students

- Since 2026 Parsa Babadi Noroozi. *TBA*. Principal supervisor (50%) with Mansooreh Zahedi and Simon Mutch. School of Computing and Information Systems, The University of Melbourne.
- Since 2025 Lantao Zhang. *Optimisation of production by selected turbine de-rating*. Principal supervisor (50%) with Michael Kirley and Kate Smith-Miles. School of Computing and Information Systems, The University of Melbourne. In collaboration with AGL Energy. **(Industry-based PhD)**
- Since 2024 Margherita Battistotti. *Multi-criteria Automatic Algorithm Configuration under Streaming Problem Instances*. Co-supervisor (30%) with Kate Smith-Miles, Manuel López-Ibañez, and Julia Handl. School of Mathematics and Statistics, The University of Melbourne; and Alliance Manchester Business School, The University of Manchester. **(Joint program)**
- Since 2023 Behzad Moradi. *Machine Learning-based Exploratory Landscape Analysis Approaches for Multi-objective Continuous Black-box Optimization Problems*. Joint supervisor (50%) with Michael Kirley. School of Computing and Information Systems, The University of Melbourne.
- Since 2022 Anthony Rasulo. *Explainable Algorithm Selection and Configuration through Instance Space*. Co-supervisor (30%) with Kate Smith-Miles, Manuel López-Ibañez, and Julia Handl. School of Mathematics and Statistics, The University of Melbourne; and Alliance Manchester Business School, The University of Manchester. **(Joint program)**
- Since 2021 Luxin Fang. *Topology estimation using smart meter measurements*. Joint supervisor (50%) with Lachlan Andrew. School of Computing and Information Systems, The University of Melbourne.

Completed

Doctoral students

- 2020-2025 Hanan Alsouly. *Algorithm Selection and Landscape Analysis for Constrained Multi-Objective Optimisation Problems*. Joint supervisor (50%) with Michael Kirley. School of Computing and Information Systems, The University of Melbourne. Currently Lecturer at Al-Imam Muhammad Ibn Saud Islamic University, Saudi Arabia.
- 2020-2024 Nicolau Andres Thio. *Multi-fidelity Expensive Black-Box Modelling: An examination of trust*. Joint supervisor (50%) with Kate Smith-Miles. School of Mathematics and Statistics, The University of Melbourne. In collaboration with Boeing Australia **(Industry-based PhD)** Currently Consultant at Endgame Analytics, Australia.
- 2019-2023 Hossein Alipour. *Empirical analysis of dynamic maximum flow problems*. Joint supervisor (50%) with Kate Smith-Miles. School of Mathematics and Statistics, The University of Melbourne. Currently Senior Technology Solutions Specialist at Telstra.
- 2017-2022 Estefania Yap. *Instance spaces for multi-objective black box optimisation*. Joint supervisor (50%) with Kate Smith-Miles. School of Mathematics and Statistics, The University of Melbourne. Currently Chief of AI Systems at Australian Systems Research Pty Ltd.

Master students

- 2025 Parsa Babadi Noroozi **(MDS)** *Anomaly detection on Satellite telemetry data*. Sole supervisor. School of Computing and Information Systems, The University of Melbourne.
- 2025 Sanjit Kumar Raj Kumar **(MCS)** *Benchmarking algorithms with a minimal test set*. Sole supervisor. School of Computing and Information Systems, The University of Melbourne.

- 2024-2025 Ashar Siddiqui (**MCS**) *Detecting similarities between datasets using kernel methods*. Sole supervisor. School of Computing and Information Systems, The University of Melbourne.
- 2024 Pan Yue (**MCS**) *Robustness of Hyperparameters in Machine Learning Surrogates for Contact Networks in Epidemic Transmission Models*. Co-supervisor (45%) with Cameron Zachreson. School of Computing and Information Systems, The University of Melbourne.
- 2024 Yin-Shao Chen (**MCS**) *Interpretable surrogates: analyzing the interpretability of machine learning surrogates of infectious disease agent-based model*. Co-supervisor (45%) with Cameron Zachreson. School of Computing and Information Systems, The University of Melbourne.
- 2022 Runqi Zhao (**MInfoTech**) *Parallel Factor Analysis of biomechanical variables of ACL-Reconstruction subjects in single-leg hop landing test*. Sole supervisor. School of Computing and Information Systems, The University of Melbourne.
- 2022 Paul Ou (**MInfoTech**) *Topological Data Analysis for Automated Algorithm Selection*. Sole supervisor. School of Computing and Information Systems, The University of Melbourne.
- 2019-2020 Hao Li (**MSc**) *Evaluation of randomized heuristic algorithm behaviour through trajectory analysis*. Sole supervisor. School of Mathematics and Statistics, The University of Melbourne.
- 2019-2020 Christopher Hadi Oetomo (**MSc**) *Analysis of online players' behaviour in a team-based video game*. Sole supervisor. School of Computing and Information Systems, The University of Melbourne.
- 2019 Zhenhao Yu (**MInfoTech**) *Testing the effectiveness of PARAFAC model in feature reduction in abnormal gait recognition*. Sole supervisor. School of Computing and Information Systems, The University of Melbourne.

Publications

Journal articles

- J1. K. Malan and **M.A. Muñoz**. *An Instance Space Analysis of Neural Network Training as a Black-Box Optimisation Problem*. Accepted to [ACM Transactions on Evolutionary Learning and Optimisation](#) (IF:4.66) in Aug-2026.
- J2. H. Avila-Choconta, L. Ruiz-Acosta, D.A. Camargo-Mayorga, and **M.A. Muñoz** (2026) *Modeling sustainability performance in manufacturing companies*. Corporate Social Responsibility and Environmental Management, 30(5)6246–6263 [\[doi\]](#) (IF:8.3)
- J3. J. Wang, L. Zhang, K. Zhang, **M.A. Muñoz**, P. Liu, X. Xue, X. Yan, Y. Yang, J. Wang, H. Sun, Q. Dai, D. Wu, and G. Qin. *Knowledge transfer across development stages for dynamic reservoir production optimization with infill wells*. Petroleum Science [\[doi\]](#) (IF:6.4)
- J4. B. Moradi, **M.A. Muñoz**, and M. Kirley (2026) *Beyond Distribution Shift: Investigating Generalisation in Automated Algorithm Selection for Single-Objective Black-Box Optimisation*. IEEE Transactions on Evolutionary Computation [\[doi\]](#) (IF:12.0)
- J5. C. Simpson, **M.A. Muñoz**, and R.J.G.B. Campello (2026) *Instance Space of Clustering Validation Measures*. Data Mining and Knowledge Discovery, 40:25 [\[doi\]](#) (IF:2.8)
- J6. C. Simpson, **M.A. Muñoz**, S. Kandanaarachchi, and R.J.G.B. Campello (2025) *ISA³ – A 3-Dimensional Expansion of Instance Space Analysis*. Machine Learning, 114:240 [\[doi\]](#) (IF:4.3)
- J7. Y.B. Güzel, K. Khare, N. Harvey, K. Dsouza, D.H. Jang, J. Chen, C.Z. Lam, and **M.A. Muñoz** (2025) *instancespace: a Python Package for Insightful Algorithm Testing through Instance Space Analysis*. SoftwareX, 31:102246 [\[doi\]](#) (IF:2.4)
- J8. B. Oldfield, S. Kandanaarachchi, Z. Xu, and **M.A. Muñoz** (2025) *An Item Response Theory-based R module for Algorithm Portfolio Analysis*. SoftwareX, 31:102239 [\[doi\]](#) (IF:2.4)
- J9. P. Sritharan, M.G. King, **M.A. Muñoz**, J.J. Heerey, M.G. Scholes, B.F. Mentiplay, A.I. Semciw, J.L. Kemp, P. Lawrenson, K.M. Crossley, and A.G. Schache (2025) *Biomechanical features of a novel step-down-and-pivot task in football players with and without hip/groin pain*. Royal Society Open Science, 12:240908 [\[doi\]](#) (IF:2.9)
- J10. A. Nikolikj, **M.A. Muñoz**, and T. Eftimov (2025) *Performance Footprints of Continuous Black-Box Optimization Algorithms: Explainable Insights into Algorithm Success and Failure*. Swarm and Evolutionary Computation, 94:101895 [\[doi\]](#) (IF:8.2)
- J11. N. Andrés-Thio, **M.A. Muñoz**, and K. Smith-Miles (2024) *Characterising harmful data sources when constructing multi-fidelity surrogate models*. Artificial Intelligence, 336:104207 [\[doi\]](#) (IF:5.1)
- J12. L. Fang, A.B. Pengwah, L. Andrew, R. Razzaghi, and **M.A. Muñoz** (2024) *Three-Phase Voltage Sensitivity Estimation and its Application to Topology Identification*. International Journal of Electrical Power and Energy Systems, 158:109949 [\[doi\]](#) (IF:5.2)

- J13. N. Andrés-Thio, **M.A. Muñoz**, and K. Smith-Miles (2024) *Methodology and challenges of surrogate modelling methods for multi-fidelity expensive black-box problems*. The ANZIAM Journal, 66(1)35–61 [\[doi\]](#)
- J14. J.L.J. Pereira, K. Smith-Miles, **M.A. Muñoz**, and A.C. Lorena (2024) *Optimal selection of benchmarking datasets for unbiased machine learning algorithm evaluation*. Data Mining and Knowledge Discovery, 38:461–500 [\[doi\]](#) (IF:5.406)
- J15. H. Alipour, **M.A. Muñoz**, and K. Smith-Miles (2024) *On the Impact of Initialisation Strategies on Maximum Flow Algorithm Performance*. Computers & Operations Research, 163:106492, [\[doi\]](#) (IF:4.6)
- J16. H. Alsouly, M. Kirley, and **M.A. Muñoz** (2023) *An Instance Space Analysis of Constrained Multi-Objective Optimization Problems*. IEEE Transactions on Evolutionary Computation, 27(5)1427–1439 [\[doi\]](#) (IF:16.497)
- J17. Neelofar, K. Smith-Miles, **M.A. Muñoz**, and A. Aleti (2023) *Instance Space Analysis of Search-Based Software Testing*. IEEE Transactions on Software Engineering, 49(4)2642–2660 [\[doi\]](#) (IF:9.322)
- J18. K. Smith-Miles, and **M.A. Muñoz** (2023) *Instance Space Analysis for Algorithm Testing: Methodology and Software Tools*. ACM Computing Surveys, 55(12:255)1–31 [\[doi\]](#) (IF:14.324)
- J19. H. Alipour, **M.A. Muñoz**, and K. Smith-Miles (2023) *Enhanced Instance Space Analysis for the Maximum Flow Problem*. European Journal of Operational Research, 304(2)411–428 [\[doi\]](#) (IF:6.363)
- J20. N. Andrés-Thio, **M.A. Muñoz**, and K. Smith-Miles (2022) *Bi-fidelity Surrogate Modelling: Showcasing the need for new test instances*. INFORMS Journal on Computing, 34(6)3007–3022 [\[doi\]](#) (Featured Article, IF:3.288)
- J21. K. Smith-Miles, and **M.A. Muñoz** (2022) *Human versus computer construction of mathematical artworks on an order-disorder aesthetic spectrum*. Journal of Mathematics and the Arts, 16(4)347–373 [\[doi\]](#)
- J22. E. Yap, **M.A. Muñoz**, and K. Smith-Miles (2022) *Informing multi-objective optimisation benchmark construction through Instance Space Analysis*. IEEE Transactions on Evolutionary Computation, 26(6)1246–1260 [\[doi\]](#) (IF:16.497)
- J23. M. Becerra-Fernández, L.E. Ruiz-Acosta, D.A. Camargo-Mayorga, and **M.A. Muñoz** (2022) *A system dynamics model for sustainable corporate strategic planning*. Production, 32:e20220011 [\[doi\]](#)
- J24. **M.A. Muñoz**, M. Kirley, and K. Smith-Miles (2022) *Analyzing randomness effects on the reliability of Landscape Analysis*. Natural Computing, 21:131–154 [\[doi\]](#) (IF:1.504)
- J25. P. Sritharan, **M.A. Muñoz**, P. Pivonka, A.L. Bryant, H. Mokhtarzadeh, and L.G. Perraton (2022) *Biomechanical markers of forward hop-landing after ACL-reconstruction: a pattern recognition approach*. Annals of Biomedical Engineering, 50(3)330–342 [\[doi\]](#) (Co-first author, IF:4.219)
- J26. E. Yap, **M.A. Muñoz**, and K. Smith-Miles (2021) *On the diversity and robustness of parameterised multi-objective test suites*. Applied Soft Computing, 110:107613 [\[doi\]](#) (IF:8.263)
- J27. S. Kandanaarachchi, N. Anantharama, and **M.A. Muñoz** (2021) *Early detection of vegetation ignition due to powerline faults*. IEEE Transactions on Power Delivery, 36(3)1324–1334 [\[doi\]](#) (IF:4.825)
- J28. K. Smith-Miles, J. Christiansen, and **M.A. Muñoz** (2021) *Revisiting Where are the Hard Knapsack Problems? via Instance Space Analysis*. Computers & Operations Research, 128:105184 [\[doi\]](#) (IF:5.159)
- J29. **M.A. Muñoz**, T. Yan, M.R. Leal, K. Smith-Miles, A.C. Lorena, G.L. Pappa, and R. Madureira Rodrigues (2021) *An Instance Space Analysis of Regression Problems*. ACM Transactions on Knowledge Discovery from Data, 15(2:28)1–25 [\[doi\]](#) (IF:4.157)
- J30. **M.A. Muñoz**, and M. Kirley (2021) *Sampling effects on algorithm selection for continuous black-box optimization*. Algorithms, 14(1)19 [\[doi\]](#) (IF:1.8)
- J31. **M.A. Muñoz**, and K. Smith-Miles (2020) *Generating new space-filling test instances for continuous black-box optimization*. Evolutionary Computation, 28(3)379–404 [\[doi\]](#) (IF:4.766)
- J32. P. Sritharan, L. Perraton, **M.A. Muñoz**, P. Pivonka, and A. Bryant (2020) *Muscular coordination of single-leg hop-landing in uninjured and ACL-reconstructed individuals*. Journal of Applied Biomechanics, 36(4)235–243 [\[doi\]](#) (IF:1.606)
- J33. M.B. Flegg, **M.A. Muñoz**, K. Smith-Miles, W.S. Yuen, J. Flegg, and J. Carroll (2020) *Parameter estimation for a point-source diffusion-decay morphogen model*. Journal of Mathematical Biology, 80:2227–2255 [\[doi\]](#) (IF:2.164)
- J34. P.D. Talagala, R.J. Hyndman, K. Smith-Miles, S. Kandanaarachchi, and **M.A. Muñoz** (2020) *Anomaly detection in streaming nonstationary temporal data*. Journal of Computational and Graphical Statistics, 29(1)13–27 [\[doi\]](#) (IF:1.884)
- J35. S. Kandanaarachchi, **M.A. Muñoz**, R.J. Hyndman, and K. Smith-Miles (2020) *On normalization and algorithm selection for unsupervised outlier detection*. Data Mining and Knowledge Discovery, 34:309–354 [\[doi\]](#) (IF:5.406)
- J36. Y. Yuan, M. Yellishetty, G.M. Mudd, **M.A. Muñoz**, S.A. Northey, and T.T. Werner (2020) *Toward dynamic evaluations of materials criticality: A systems framework applied to Platinum*. Resources, Conservation & Recycling, 152:104532 [\[doi\]](#) (IF:13.716)
- J37. Y. Yuan, M. Yellishetty, **M.A. Muñoz**, and S.A. Northey (2019) *Towards a dynamic evaluation of minerals criticality: Introducing the framework of criticality systems*. Journal of Industrial Ecology, 23(5)1264–1277 [\[doi\]](#) (IF:4.9)

- J38. **M.A. Muñoz**, L. Villanova, D. Baatar, and K. Smith-Miles (2018) *Instance spaces for machine learning classification*. Machine Learning, 107(1)109–147 [doi] (IF:5.414)
- J39. **M.A. Muñoz**, and K.A. Smith-Miles (2017) *Performance analysis of continuous black-box optimization algorithms via footprints in instance space*. Evolutionary Computation, 25(4)529–554 [doi] (IF:4.766)
- J40. **M.A. Muñoz**, Y. Sun, M. Kirley, and S.K. Halgamuge (2015) *Algorithm selection for black-box continuous optimization problems: a survey on methods and challenges*. Information Sciences, 317:224–245 [doi] (IF:8.233)
- J41. **M.A. Muñoz**, M. Kirley, and S.K. Halgamuge (2015) *Exploratory Landscape Analysis of Continuous Space Optimization Problems using Information Content*. IEEE Transactions on Evolutionary Computation, 19(1)74–87 [doi] (IF:16.497)
- J42. H. Mokhtarzadeh, L. Perraton, L. Fok, **M.A. Muñoz**, R. Clark, P. Pivonka, and A.L. Bryant (2014) *A comparison of optimization methods and knee joint degrees of freedom on muscle force predictions during single-leg hop landings*. Journal of Biomechanics, 47(12)2863–2868 [doi] (IF:2.789)
- J43. **M.A. Muñoz**, J.A. López, and E.F. Caicedo (2009) *An Artificial Beehive for Continuous Optimization*. International Journal of Intelligent Systems, 24(11)1080–1093 [doi] (IF:8.993)
- J44. **M.A. Muñoz**, J.A. López, and E.F. Caicedo (2008) *Inteligencia de enjambres: sociedades para la solución de problemas (una revisión) | Swarm intelligence: problem-solving societies (a review)* Revista Ingeniería e Investigación, 28(2)119–130 [doi] (In Spanish)
- J45. **M.A. Muñoz**, J.A. López, and E.F. Caicedo (2007) *Optimización por Colonia de Hormigas para la Asignación Dinámica de Recursos en una Plataforma de Experimentación de Temperatura Multizona | Ant Colony Optimization for Dynamical Resource Allocation in a Multizone Temperature Experimentation Platform*. IEEE Latin America Transactions, 5(2)82–87 [doi] (In Spanish)

Scholarly books

- B1. E. Caicedo, J.A. López, and **M.A. Muñoz** (2010) *Control Inteligente | Intelligent Control*. Programa Editorial Universidad del Valle, ISBN:978-958-670-962-0 (In Spanish)

Scholarly book chapters

- C1. **M.A. Muñoz**, A. Muñoz del Castillo, S. Shokravi, and J.A. Jiménez Toledo (2017) *TITAN: una herramienta para analizar Ambientes Virtuales de Aprendizaje – Diseño preliminar | TITAN: An analysis tool for Virtual Learning Environments - Preliminary Design*. In Tecnología, innovación e investigación en los procesos de enseñanza-aprendizaje. pp. 927–935. Octaedro (In Spanish)
- C2. **M.A. Muñoz**, M. Kirley, and S.K. Halgamuge (2012) *The Algorithm Selection Problem on the Continuous Optimization Domain*. In: Moewes, C., Nürnberger, A. (eds) Computational Intelligence in Intelligent Data Analysis. Studies in Computational Intelligence, vol 445. Springer, Berlin, Heidelberg. [doi]

Refereed conference papers

- P1. M. Battistotti, M. López-Ibañez, K. Smith-Miles, J. Handl, and **M.A. Muñoz** (2026) *Constructing Streams of Optimization Instances for Benchmarking AAS and AAC in Streaming Scenarios*. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '26), pp. 4–12 [doi]
- P2. A. Rasulo, M. López-Ibañez, K. Smith-Miles, J. Handl, and **M.A. Muñoz** (2026) *An Efficient Hybrid Racing Method for Portfolio Configuration*. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '26), pp. 701–708 [doi]
- P3. B. Moradi, **M.A. Muñoz**, and M. Kirley (2026) *Generalisation of Automated Algorithm Selection in Black-Box Optimisation: The Role of Algorithm Portfolio and Learning Model*. In: García-Sánchez, P., Díaz Álvarez, J., Murphy, A. (eds) Applications of Evolutionary Computation. EvoApplications 2026. Lecture Notes in Computer Science, vol 16524, pp. 333–349. Springer, Cham. [doi]
- P4. D. Notice, H. Soleimani, N.G. Pavlidis, A. Kheiri, and **M.A. Muñoz** (2025) *Instance Space Analysis of the Capacitated Vehicle Routing Problem with Mixture Discriminant Analysis*. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '25), pp. 1172–1180 [doi]
- P5. K. Malan, and **M.A. Muñoz** (2025) *Why We Should be Benchmarking Evolutionary Algorithms on Neural Network Training Tasks*. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '25), pp. 30–38 [doi] (Best Paper Nomination)
- P6. A. Nikolikj, **M.A. Muñoz**, E. Tuba, and T. Eftimov (2025) *Tracing the Interactions of Modular CMA-ES Configurations Across Problem Landscapes*. In Proceedings of the 2025 IEEE Congress on Evolutionary Computation (CEC 2025) [doi]

- P7. M. Gallagher, and **M.A. Muñoz** (2024) *Towards an Improved Understanding of Features for More Interpretable Landscape Analysis*. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24), pp. 135–138 [\[doi\]](#)
- P8. B. Moradi, M. Kirley, and **M.A. Muñoz** (2024) *Sensitivity Analysis of Surrogate-assisted Bilevel Optimisation*. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24), pp. 411–414 [\[doi\]](#)
- P9. A. Rasulo, K. Smith-Miles, **M.A. Muñoz**, J. Handl, and M. López-Ibañez (2024) *Extending Instance Space Analysis to Algorithm Configuration Spaces*. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24), pp. 147–150 [\[doi\]](#)
- P10. H. Alsouly, M. Kirley, and **M.A. Muñoz** (2024) *Online Per-Instance Algorithm Selection for Constrained Multi-Objective Optimization Problems*. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24), pp. 559–562 [\[doi\]](#)
- P11. H. Alsouly, M. Kirley, and **M.A. Muñoz** (2023) *Dynamic Landscape Analysis for Constrained Multiobjective Optimization Problems*. In Proceedings of the Australasian Joint Conference on Artificial Intelligence (AI '23), pp. 429–441 [\[doi\]](#).
- P12. A. Nikolikj, S. Dzeroski, **M.A. Muñoz**, C. Doerr, P. Korosec, and T. Eftimov (2023) *Algorithm Instance Footprint: Separating Easily Solvable and Challenging Problem Instances*. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '23), pp. 529–537 [\[doi\]](#).
- P13. **M.A. Muñoz** (2022) *Examining Algorithm Behavior using Recurrence Quantification and Landscape Analyses*. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '22), pp. 1658–1665 [\[doi\]](#)
- P14. **M.A. Muñoz**, H. Soleimani, and S. Kandanaarachchi (2022) *Benchmarking Algorithm Portfolio Construction Methods*. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '22), pp. 499–502 [\[doi\]](#)
- P15. D.X. Ramos, J.A. Jiménez Toledo, A. Muñoz del Castillo, L.C. Acosta Huertas, E. Herrera, and **M.A. Muñoz** (2020) *Computational Thinking for Teacher Training: A Systematic Review of Literature*. In Proceedings of 18th LACCEI International Multi-Conference for Engineering, Education and Technology [\[doi\]](#) **(In Spanish)**
- P16. E. Yap, **M.A. Muñoz**, K. Smith-Miles, and A. Liefooghe (2020) *Instance Space Analysis of Combinatorial Multi-objective Optimization Problems*. In Proceedings of the 2020 IEEE Congress on Evolutionary Computation [\[doi\]](#)
- P17. S. Kandanaarachchi, **M.A. Muñoz**, and K. Smith-Miles (2019) *Instance space analysis for unsupervised outlier detection*. In Proceedings of the 1st Workshop on Evaluation and Experimental Design in Data Mining and Machine Learning, co-located with SIAM International Conference on Data Mining (SDM 2019), pp. 32–41 [\[html\]](#)
- P18. A. Muñoz del Castillo, **M.A. Muñoz**, L.C. Acosta Huertas, E. Herrera, J.A. Jiménez Toledo, and D.X. Ramos (2019) *Developing a teacher training curriculum including Computational Thinking skills: Early advances on a study focused on Colombia*. In Proceedings of 2019 XIV Latin American Conference on Learning Technologies (LACLO), pp. 8–11 [\[doi\]](#)
- P19. **M.A. Muñoz**, and K.A. Smith-Miles (2017) *Generating custom classification datasets by targeting the instance space*. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '17), pp. 1582–1588 [\[doi\]](#)
- P20. **M.A. Muñoz**, and K.A. Smith-Miles (2017) *Non-parametric model of the space of continuous black-box optimization problems*. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '17), pp. 175–176 [\[doi\]](#)
- P21. **M.A. Muñoz**, and M. Kirley (2016) *ICARUS: Identification of Complementary algorithms by Uncovered Sets*. In Proceedings of the 2016 IEEE Congress on Evolutionary Computation, pp. 2427–2432 [\[doi\]](#)
- P22. **M.A. Muñoz**, and K.A. Smith-Miles (2015) *Effects of function translation and dimensionality reduction on landscape analysis*. In Proceedings of the 2015 IEEE Congress on Evolutionary Computation, pp. 1336–1342 [\[doi\]](#)
- P23. Y. Sun, M. Kirley, S.K. Halgamuge, and **M.A. Muñoz** (2014) *On the selection of fitness landscape analysis metrics for continuous optimization problems*. In Proceedings of the 7th International Conference on Information and Automation for Sustainability [\[doi\]](#)
- P24. **M.A. Muñoz**, M. Kirley, and S.K. Halgamuge (2012) *A Meta-Learning prediction model of algorithm performance for continuous optimization problems*. In Proceedings of the 12th International Conference on Parallel Problem Solving from Nature (PPSN XII), Lecture Notes in Computer Science, v. 7491, pp. 226–235 [\[doi\]](#)
- P25. **M.A. Muñoz**, M. Kirley, and S.K. Halgamuge (2012) *Landscape characterization of numerical optimization problems using biased scattered data*. In Proceedings of the 2012 IEEE Congress on Evolutionary Computation, pp. 2162–2169 [\[doi\]](#)
- P26. **M.A. Muñoz**, S.K. Halgamuge, W. Alfonso, and E.F. Caicedo (2010) *Simplifying the Bacteria Foraging Algorithm*. In Proceedings of the 2010 IEEE Congress on Evolutionary Computation, pp. 4095–4101 [\[doi\]](#)

- P27. **M.A. Muñoz**, J.A. López, and E.F. Caicedo (2008). *Self-Adaptive Bacteria Swarm for Optimization*. In Proceedings of the 2008 Electronics, Robotics and Automotive Mechanics Conference (CERMA '08), pp. 45–49 [doi]
- P28. **M.A. Muñoz**, J.A. López, and E.F. Caicedo (2007) *Bacteria Swarm Foraging Optimization for Dynamical Resource Allocation in a Multizone Temperature Experimentation Platform*. In Proceedings of the 2007 International Fuzzy Systems Association Congress (IFSA 2007), Analysis and Design of Intelligent Systems using Soft Computing Techniques. Advances in Soft Computing, v. 41, pp. 427–435 [doi]
- P29. **M.A. Muñoz**, J.A. López, and E.F. Caicedo (2006) *Ant Colony Optimization for Dynamical Resource Allocation in a Multizone Temperature Experimentation Platform*. In Proceedings of the 2006 Electronics, Robotics and Automotive Mechanics Conference (CERMA '06), pp. 137–142 [doi]
- P30. **M.A. Muñoz**, J.A. López, and E.F. Caicedo (2005) *Implementation of a Distributed Control Experimentation Platform*. In Proceedings of the 2005 International Conference on Industrial Electronics and Control Applications (ICIECA 2005) [doi]

Work under review or revision

- R1. B. Moradi, **M.A. Muñoz**, and M. Kirley. *Landscape-Aware Automated Algorithm Selection for Black-Box Bilevel Optimisation*. Submitted to [IEEE Transactions on Evolutionary Computation](#) on Jul-2026 (IF:12.0)
- R2. **M.A. Muñoz**. *Bounding the Drift Gap via Landscape Similarity in Black-Box Optimisation*. Submitted to [IEEE Transactions on Evolutionary Computation](#) on Jul-2026 (IF:12.0)
- R3. **M.A. Muñoz**. *Modelling Black-Box Optimisation Landscapes as Probability Distributions: A Maximum Mean Discrepancy Approach to Instance Space Embeddings*. Submitted to [Journal of Artificial Intelligence Research](#) (IF:5.94) First round of reviews received in Jul-2026.
- R4. A. Rasulo, K. Smith-Miles, **M.A. Muñoz**, J. Handl, and M. López-Ibañez. *Algorithm Insights via a Flexible Instance-Configuration Space Projection*. Submitted to [Journal of Artificial Intelligence Research](#) on Jan-2026 (IF:5.94)
- R5. M. Yellishetty, Y. Yuan, T.T. Werner, S.A. Northey, W.Q. Chen, **M.A. Muñoz**, and G. Mudd. *Assessing Platinum and Palladium Price Dynamics through the Lens of Metal Criticality*. Submitted to [Mineral Economics](#) (IF:3.0) First round of reviews received on Jan-2026.